

We want to make sure that throughout the execution of the algorithm, $\Phi(I) \leq \mu \cdot w(I)$. Before any interval is accepted through replacement, intervals in the solution could have only been directly charged through rejected optimal intervals, and because of (1), and the fact that $\lambda > 1$, our desired bound holds. We now consider all the cases of an interval being accepted through replacement.

Case 1: I is an optimal interval and it is accepted through the predictions rule. In this case we have that $DC(I) = w(I)$, and we need to look at $TC(I)$. Let L_c and R_c denote the intervals (if any) that I is partially conflicting with on the left and on the right respectively, and let M_c denote the set of intervals that I subsumes. We know that all of these conflicting intervals are not optimal, and they were accepted through the algorithm's main rule. First, notice that for all $J \in M_c$, $DC(J) = 0$, and $\Phi(J) = TC(J) \leq \frac{\mu}{\lambda} \cdot w(J)$. Moreover, L_c and R_c had not yet been directly charged by a partially conflicting optimal interval on one side, and therefore we have that $\Phi(L_c) \leq \frac{\mu}{\lambda} \cdot w(L_c) + 2w(L_c)$, and similarly $\Phi(R_c) \leq \frac{\mu}{\lambda} \cdot w(R_c) + 2w(R_c)$. Putting everything together:

$$\begin{aligned} TC(I) &= \sum_{J \in M_c} \Phi(J) + \Phi(L_c) + \Phi(R_c) \\ &\leq \frac{\mu}{\lambda} \cdot w_c + 2(w(L_c) + w(R_c)) \\ &\leq \left(\frac{\mu}{\lambda} + 2\right) w_c \\ &\leq \left(\frac{\mu}{\lambda} + 2\right) w(I) \end{aligned}$$

The last inequality being true from the fact that the main predictions rule is satisfied. Given also that $DC(I) = w(I)$, we get that $\Phi(I) \leq \left(\frac{\mu}{\lambda} + 2\right)w(I) + w(I) = \left(\frac{\mu}{\lambda} + 3\right)w(I)$. With our choice of μ , we have:

$$\begin{aligned} \Phi(I) &\leq \left(\frac{3\lambda}{\lambda-1} + 3\right) w(I) \\ &= \left(\frac{3\lambda}{\lambda-1}\right) w(I) \end{aligned}$$

Case 2: I is an optimal interval and it is accepted through the algorithm's main rule. This is similar to case 1, with $DC(I) = w(I)$ and $w(I) \geq \lambda \cdot w_c$. The same analysis gives us $TC(I) \leq \left(\frac{\mu}{\lambda} + 2\right) \frac{w(I)}{\lambda}$, and because $\lambda > 1$, the same bound holds.

Case 3: I is not an optimal interval and it is accepted through the algorithm's main rule. In this case we have that $DC(I) \leq 3w(I)$, and we get that

$$\begin{aligned} \Phi(I) &\leq \sum_{J \in I_s} \Phi(J) + 3w(I) \\ &\leq \frac{\mu}{\lambda} \cdot w(I) + 3w(I) \\ &= \left(\frac{3\lambda}{\lambda-1}\right) w(I) \end{aligned}$$

In conclusion, we have that throughout the execution of the algorithm, for $I \in ALG$, $\Phi(I) \leq \frac{3\lambda}{\lambda-1} w(I)$, and therefore $\frac{OPT}{ALG} \leq \frac{3\lambda}{\lambda-1}$. $\square$

We see that as $\lambda \rightarrow \infty$, the algorithm's consistency goes to 3. We now look at the robustness of algorithm Revoke-Proportional.

THEOREM 4.11. *Algorithm Revoke-Proportional is $\frac{4\lambda^2+2\lambda}{\lambda-1}$ -robust.*

PROOF. The argument is similar to the proof of Theorem 4.10. Both *direct*, and *transfer* charging work the same way as before. Let $\mu = \frac{2\lambda^2+3\lambda+1}{\lambda-1}$, and $\delta = 2\lambda + 1$. We will show that that for every $I \in ALG$, $\Phi(I) \leq (\mu + \delta) \cdot w(I) = \frac{4\lambda^2+2\lambda}{\lambda-1} w(I)$.

Notice first that the upper bound on direct charging is not as good as before. More precisely, with I_o being a newly arrived optimal interval that will be rejected and I_s being its conflicting intervals currently in the solution, we have that for every $J \in I_s$, $DC_{I_o}(J) = w(I_o) \frac{w(J)}{w_c} \leq \lambda \cdot w(J)$. More generally, $DC_{I_o}(J) \leq \min\{w(I_o), \lambda \cdot w(J)\}$. As before, given the three different possible types of conflicts, we have that:

$$DC(I) \leq (2\lambda + 1)w(I) \quad (2)$$

We can now bound the total amount of charge on every interval in the algorithm's solution, throughout its execution. Before any replacement happens, the bound $\Phi(I) \leq (\mu + \delta) \cdot w(I)$ holds trivially.

Case 1: Interval I is accepted through the algorithm's main rule. We get that:

$$\begin{aligned} \Phi(I) &\leq (\mu + \delta) \cdot w_c + (2\lambda + 1) \cdot w(I) \\ &\leq (\mu + \delta) \cdot \frac{w(I)}{\lambda} + (2\lambda + 1) \cdot w(I) \\ &= \left(\frac{\mu + \delta}{\lambda} + 2\lambda + 1\right) w(I) \\ &= \left(\frac{\frac{4\lambda^2+2\lambda}{\lambda-1} + 2\lambda^2 + \lambda}{\lambda}\right) w(I) \\ &= \mu \cdot w(I) \end{aligned}$$

Case 2: Interval I is accepted through the algorithm's predictions rule. Notice that in this case, all conflicting intervals must have been accepted through the main rule, and not the predictions rule. Because of this, as we showed in case 1, for every $J \in I_s$, it holds that $\Phi(J) \leq \mu \cdot w(J)$. This helps us bound the amount of transfer charge to interval I .

$$\begin{aligned} \Phi(I) &\leq \mu \cdot w_c + (2\lambda + 1) \cdot w(I) \\ &\leq \mu \cdot w(I) + (2\lambda + 1) \cdot w(I) \\ &= (\mu + \delta) \cdot w(I) \end{aligned}$$

To summarize, we have shown that in the worst case, $\Phi(I) \leq (\mu + \delta) \cdot w(I)$ for every $I \in ALG$. This concludes the proof. $\square$

We note that for $\lambda > \frac{2+\sqrt{5}}{\sqrt{5}-1} \approx 3.42$, the consistency of our algorithm is already better than $2\phi + 1$, and 22.15-robust. We have shown we can get consistency better than the online bound of LR, while maintaining bounded robustness. We believe further improvement on the bounds of Revoke-Proportional is possible, with an analysis that looks more closely at the dependence between direct and transfer charging.

One may also be able to further improve the algorithm by accepting

Figure 6: NASA-iPSC dataset.

(a) Unit & Irrevocable, (b) Unit & Revoking, (c) Proportional & Irrevocable, (d) Proportional & Revoking

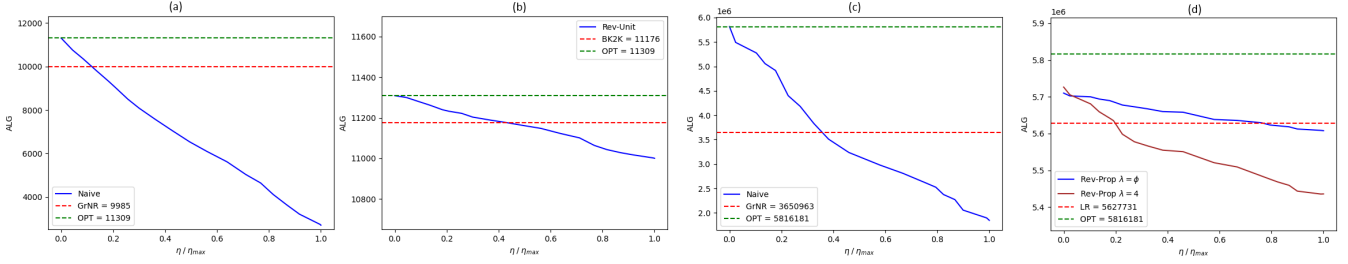
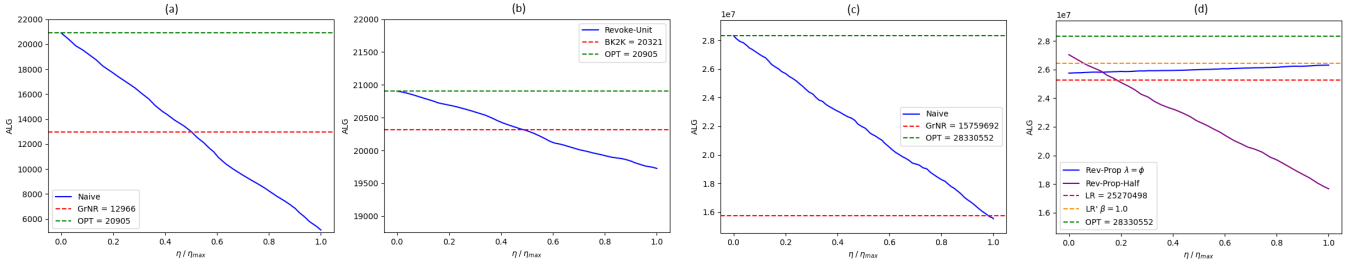


Figure 7: CTC-SP2 dataset.



an interval that is not as big as the sum of its conflicts, making the algorithm a -increasing with $a < 1$. This would relax the predictions rule further, and make the algorithm more prone to bad choices caused by misleading predictions. In our experiments, we briefly discuss one such algorithm, which we call Revoke-Prop-Half, and which can accept a supposedly optimal interval even if it is half as big as its conflicts.

5 EXPERIMENTAL RESULTS

We use real-world data from scheduling jobs on parallel machines³ to test our algorithms. More information on the handling of these datasets can be found in a study by Feitelson et al. [17]. We focus on two datasets, NASA-iPSC (18,239 jobs) and CTC-SP2 (77,222 jobs). As is usually the case, the performance of algorithms is much better than their worst-case bounds. For every algorithm we average its performance over random permutations of the input instance, for multiple error values. The y axis values for proportional weights are expressed in scientific notation. We note that algorithm GrNR refers to a greedy algorithm without revoking, a very natural algorithm to compare our Naive algorithm against. All other algorithms have been mentioned earlier in the paper. Our experimental results are in line with our intuition, with the predictions algorithms outperforming predictionless algorithms for some values of the error, even when they are not 1-consistent. Especially in the setting of revocable acceptances, even with half of the max possible error, our predictions algorithms perform just as well as their purely online

counterparts. In the CTC dataset (figure 7) this is always the case, with the Naive algorithm outperforming GrNR for nearly all values of error. In the case of proportional weights with revoking in figure 6, it is noteworthy that the variant of Revoke-Proportional with $\lambda = 4$, outperforms the $\lambda = \phi$ variant for some small values of error, but its performance degrades faster. This further validates the notion that the bigger the λ , the more the algorithm follows the predictions.

We also have to address the seemingly abnormal behavior of algorithm Revoke-Proportional in figure 7(d). As the error increases, so does the performance of Revoke-Proportional, which is counterintuitive and dissimilar to the corresponding plot of figure 6. This is because of the underlying structure of the CTC-SP2 dataset, on which *greedy* algorithms perform exceptionally well. We showcase this by having included algorithm LR' with $\beta = 1$, the algorithm that accepts a new interval if it is at least as big as everything it conflicts with. As the error increases, a larger number of intervals can be accepted through this clearly beneficial, relaxed predictions rule, which helps explain the improved performance. We also contrast this with algorithm Rev-Prop-Half, which uses a modified predictions rule, that can accept supposedly optimal intervals that are half the weight of their conflicts. This makes the algorithm more sensitive to the predictions, and its performance falls in line with what we would expect.

In conclusion, algorithms for interval selection can greatly benefit from utilizing imperfect predictions, and remain robust even in the presence of high error.

³<https://www.cs.huji.ac.il/labs/parallel/workload/>

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